

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15D.3

CARBOXYLIC ACID DERIVATIVES: POLYESTERS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15D.3: Carboxylic Acid Derivatives: Polyesters
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15D.3

Q1. 1. (a) Terylene (PET) is a condensation polymer formed from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. Draw the structural formula of the repeat unit of Terylene.

[2 Marks]

WCH14/01/Jun21/Q7(a)

Q2. (b) Explain why polyesters are biodegradable whereas addition polymers like poly(ethene) are not.

[3 Marks]

WCH14/01/Jun21/Q7(b)

SECTION B: WORKED MARK SCHEME — 15D.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Terylene (PET) is a condensation polymer formed from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. Draw the structural formula of the repeat unit of Terylene. 1. Correct ester linkage $\text{-O-CO-C}_6\text{H}_4\text{-CO-O-CH}_2\text{-CH}_2\text{-}$. 2. Open continuation bonds at both ends of the repeat unit.	[2]
Q2	Q2: (b) Explain why polyesters are biodegradable whereas addition polymers like poly(ethene) are not. 1. Polyesters contain polar ester links (-CO-O-) susceptible to hydrolysis by microorganisms/water. 2. Poly(ethene) has strong non-polar C-C and C-H single bonds resistant to chemical and enzymatic attack.	[3]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15D.3

FAQ #1: How do I identify the monomers from a polymer repeat unit?

Category: Polymer Analysis

COMMON EXAMINER TRAP / MISTAKE:

Adding water in the wrong places when breaking ester links.

MODEL ANSWER / EXAMINER EXPECTATION:

Break the C–O ester single bond in the repeat unit. Add –OH to the carbonyl C=O group to regenerate the carboxylic acid, and add –H to the oxygen atom to regenerate the alcohol.